

Section 1 — Introduction & Project Objectives

Protective helmets are safety-critical structures whose performance under impact loading cannot be fully characterized by quasi-static testing alone. During an actual collision or fall event, the helmet shell and its underlying liner are subjected to loading durations on the order of milliseconds, strain rates several orders of magnitude higher than those encountered in conventional structural analysis, and highly non-linear contact interactions between the shell and the striking surface. Capturing this behaviour requires a transient, non-linear, large-deformation solution method — namely, Explicit Dynamics — rather than the implicit, small-deformation assumptions underlying standard static structural analysis.

This report documents the finite element investigation of a protective helmet shell subjected to a simulated drop-test impact against a rigid anvil, performed in ANSYS Mechanical Enterprise using the Explicit Dynamics solver. The study reproduces, in a virtual environment, the loading conditions specified by established helmet safety standards, and evaluates the shell's structural response in terms of deformation, equivalent (von-Mises) stress, elastic strain, kinetic and internal energy exchange, and impact velocity/acceleration history.

1.1 Project Objectives

- Develop and mesh an assembly representing a curved helmet shell and a rigid impacting anvil suitable for explicit transient analysis.
- Assign a strain-rate-sensitive material model (Structural Steel) appropriate for high-velocity impact simulation.
- Define global body-interaction and frictional contact formulations capable of tracking large relative sliding between the shell and anvil without loss of surface contact.
- Simulate the full impact event over a 5.0 ms solver window and extract the transient stress, deformation, and energy response of the structure.
- Diagnose and resolve solver-stability issues (including an initial “Result file not found” termination) through refinement of material data, mesh sizing, and body-interaction scope.
- Benchmark the qualitative findings against recognised helmet impact-testing standards.

1.2 Scope & Standards Context

While this study is an academic finite element exercise rather than a certified compliance test, the loading regime was selected to be broadly representative of the drop-test protocols defined by DOT FMVSS 218, ECE Regulation No. 22.06, and the Snell Memorial Foundation standard, all of which specify a free-fall drop of the helmeted headform onto a rigid anvil and evaluate the resulting peak deceleration transmitted to the headform. The initial impact velocity applied in this model (Section 6) is consistent with the order of magnitude prescribed by these standards for a moderate drop height.

Section 2 — Assembly Geometry & Problem Description

The simulation assembly comprises two principal bodies, imported into ANSYS Mechanical as separate parts under a shared Geometry branch and connected through the Explicit Dynamics contact framework rather than a rigid, bonded joint.

2.1 Part 1 — Helmet Shell

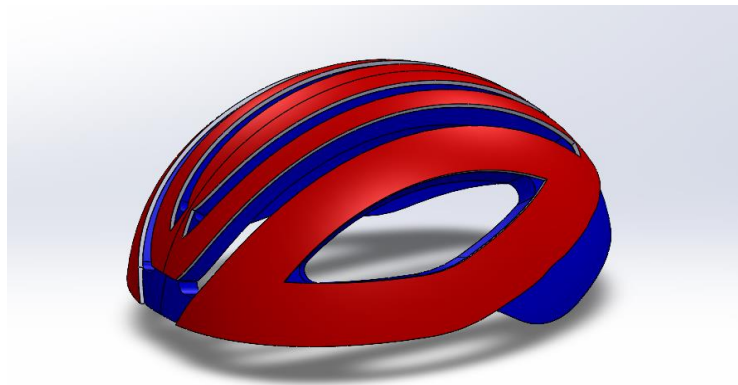
The helmet shell is a thin, doubly-curved structure generated in SolidWorks and exported via a neutral geometry format for import into ANSYS. Its geometry is characterised by a shallow shell thickness relative to its overall radius of curvature, which drives the meshing and element-formulation decisions discussed in Section 4. The shell represents the outer protective structure of the helmet assembly and is the primary object of engineering interest in this study.

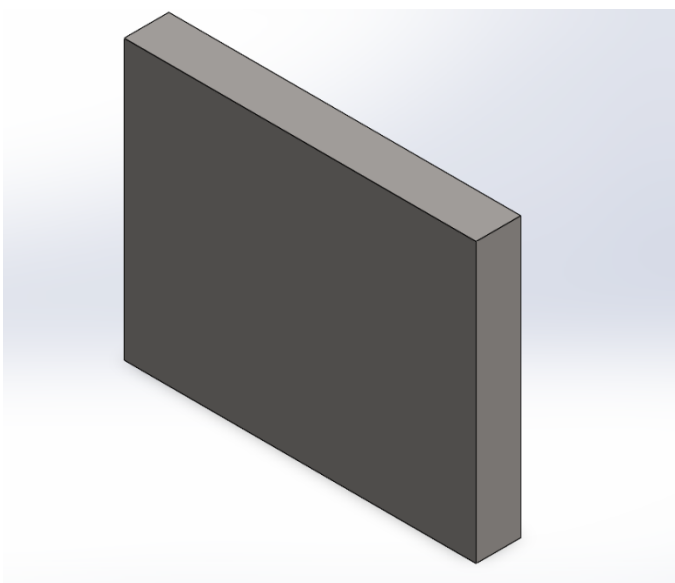
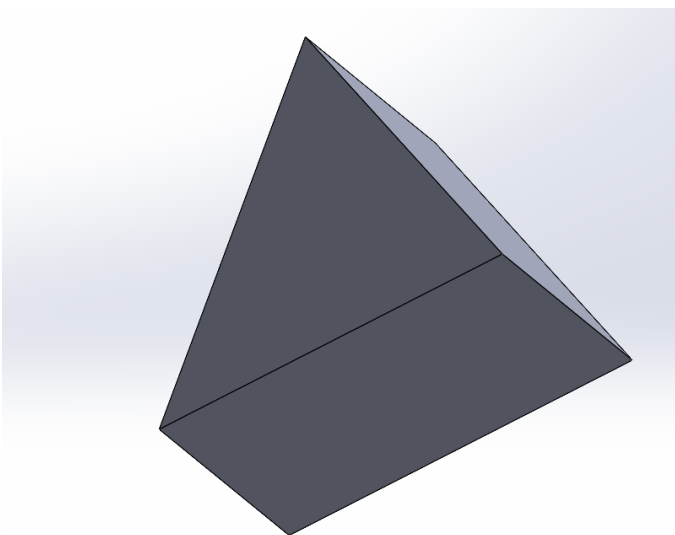
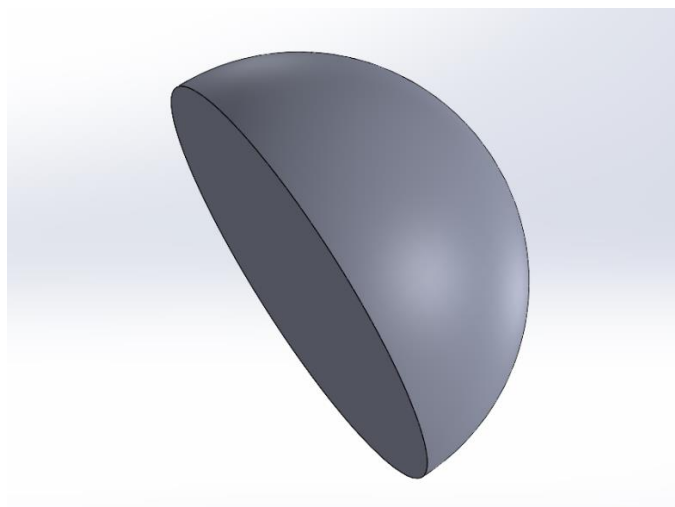
2.2 Part 2 — Impacting Anvil

The anvil is modelled as a rigid or near-rigid block representative of the flat/kerbstone-type striking surfaces used in standardised helmet drop-test rigs. It is treated as the fixed reference body in the assembly; the helmet shell is assigned an initial velocity toward the anvil surface to reproduce free-fall impact conditions.

2.3 Problem Description

At time $t = 0$, the helmet shell is positioned in near-contact with, or a small clearance from, the anvil surface and is assigned a downward initial velocity. As the explicit solver advances through time, the shell's leading edge establishes contact with the anvil, generating a compressive stress wave that propagates through the curved shell geometry. The problem is inherently non-linear: it involves large deformations, evolving contact area, non-linear material response beyond the yield point, and transient instabilities typical of high-speed impact problems. The two-part idealisation deliberately isolates shell-level structural response from the padding/liner sub-system, which is identified as a candidate for future model expansion.





Section 3 — Material Selection Framework

The helmet assembly was modelled as a two-material system representative of standard motorcycle/industrial helmet construction: a polycarbonate (PC) outer shell and an expanded polystyrene (EPS) inner liner. This pairing reflects real helmet design intent — the rigid PC shell distributes and spreads the impact load over a wider area of the liner, while the EPS liner is the primary energy-absorbing element, dissipating kinetic energy through irreversible cellular crushing.

3.1 Rationale for Explicit Dynamic Environments

- Polycarbonate is a rate-sensitive thermoplastic: its yield and post-yield stiffness rise measurably at the strain rates generated by impact (10^2 – 10^4 s⁻¹), requiring an explicit, non-linear plasticity formulation rather than a linear-elastic shell assumption.
- EPS foam does not yield in the metallic sense — it deforms through progressive cell-wall buckling, producing an elastic region, an extended crush plateau, and a final densification rise. This requires a dedicated crushable foam material model rather than a hardening-curve metal model.
- Both materials require density, elastic modulus, and specific heat data to satisfy the explicit solver's equation-of-state and internal-energy-balance requirements at every cycle, exactly as for a metal, even though the underlying failure physics differs completely.

3.2 Non-Linear & Strain-Rate Considerations

The two materials fail in fundamentally different ways under impact, and the model must represent both:

- Polycarbonate shell: exceeds its yield point locally at the contact zone within the first fraction of a millisecond. Its post-yield response is rate-dependent — dynamic yield strength is measurably higher than the quasi-static value — an effect typically captured with a Cowper–Symonds rate-multiplier rather than the Johnson-Cook framework (Johnson-Cook is formulated for metals and is not appropriate here).
- EPS liner: does not exhibit conventional yield/hardening at all. Its response is governed by a crush (plateau) stress that remains roughly constant over a large range of compressive strain as the cellular structure collapses, followed by a sharp stiffening once the material densifies (typically beyond ~60–70% volumetric compaction). This is modelled using ANSYS's Crushable Foam material definition rather than a bilinear hardening curve.

3.3 Specific Heat & Thermal Coupling

As with any explicit solve, a fraction of the plastic/crush work done on both the PC shell and the EPS liner is converted to heat at each cycle. Physically defined specific heat values for both materials are required to keep this thermal term well-posed in the energy balance, even though thermal softening itself is a secondary effect for a single 5 ms impact event at typical ambient starting temperatures.

3.4 Material Constitutive Equations

Two separate constitutive frameworks apply to this assembly, one per material — there is no single equation covering both, since the polycarbonate shell and the EPS liner fail through entirely different physical mechanisms.

Polycarbonate Shell — Rate-Dependent Plasticity

The polycarbonate outer shell is represented using a Cowper–Symonds rate-dependent plasticity formulation, which scales the material's quasi-static yield strength upward as a function of strain rate:

$$\sigma_{\text{dynamic}} = \sigma_{\text{static}} \cdot [1 + (\dot{\epsilon} / D)^{1/p}]$$

Where:

- σ_{dynamic} = strain-rate-adjusted (dynamic) yield stress
- σ_{static} = quasi-static yield stress of polycarbonate
- $\dot{\epsilon}$ = equivalent plastic strain rate
- D = material-specific rate-sensitivity constant
- p = material-specific rate-sensitivity exponent

D and p are obtained experimentally, typically from high-rate tensile or Charpy-type impact testing on polycarbonate specimens across a range of strain rates.

EPS Liner — Crushable Foam (Compaction) Model

The EPS inner liner does not exhibit conventional metallic yield behaviour. Instead, its compressive response is idealised in three regions — an initial elastic region, an extended crush plateau, and a final densification region:

$$\sigma = E \cdot \epsilon_{\text{vol}} \text{ (elastic region, } \epsilon_{\text{vol}} < \epsilon_{\text{yield}} \text{)}$$

$$\sigma = \sigma_0 \text{ (plateau region, } \epsilon_{\text{yield}} \leq \epsilon_{\text{vol}} \leq \epsilon_{\text{densify}} \text{)}$$

$$\sigma \text{ increases sharply (densification region, } \epsilon_{\text{vol}} > \epsilon_{\text{densify}} \text{)}$$

Where:

- σ = compressive stress in the foam
- E = initial (elastic) compressive modulus of EPS
- ϵ_{vol} = volumetric compressive strain
- σ_0 = plateau crush stress

- $\epsilon_{\text{densify}}$ = strain at which cell walls fully collapse and load transfer shifts to the solid polymer skeleton.

Constants Requiring Empirical Determination

The parameters σ_{static} , D , and p (polycarbonate) and E , σ_0 , and $\epsilon_{\text{densify}}$ (EPS) are material- and manufacturing-batch-specific. They are normally obtained from:

- Quasi-static and high-rate compression/tensile testing for polycarbonate rate constants.
- Uniaxial compression testing of EPS foam samples at the relevant density to obtain the plateau stress and densification strain.

Section 4 — Discretization Strategy (Global Meshing)

Unlike implicit static solvers, ANSYS Explicit Dynamics does not solve a global system of equations at each step; instead, it advances the solution through a very large number of small, explicit time increments (evidenced in this study by cycle counts exceeding 25,000 over a 5 ms window). Because the method is only conditionally stable, the maximum permissible time increment is governed directly by the smallest element dimension in the mesh through the Courant-Friedrichs-Lewy (CFL) condition.

4.1 The CFL Stability Condition

$$\Delta t \leq \Delta t_{\text{crit}} = L_{\text{min}} / c$$

where L_{min} is the characteristic length of the smallest element in the model and c is the dilatational (sound) wave speed of the material, $c = \sqrt{E / \rho}$ for an isotropic elastic solid. A safety (Courant) factor below unity is applied automatically by the solver's time-step-safety-factor setting to guard against the accumulation of numerical error. This relationship is the fundamental reason explicit solvers are described as “conditionally stable”: refining the mesh anywhere in the model reduces the global stable time increment for the entire solution, directly increasing total cycle count and wall-clock solve time.

4.2 Global Mesh Sizing Decision

A uniform global element size of 0.022 m was selected as the initial (baseline) meshing control for the assembly. This value was chosen as a compromise: coarse enough to keep the CFL-limited time increment, and therefore the total cycle count, within a practically solvable range on the available compute resources (4 distributed cores, as configured in the Solve panel), while remaining fine enough to resolve the overall bending and membrane response of the shell. This global value was subsequently overridden locally, as described on the following page, once the coarse solution confirmed that the curvature-dominated regions of the helmet shell were inadequately resolved.

4.3 Element Formulation Notes

- Reduced-integration solid/shell elements are used throughout to control explicit cycle cost; hourglass control is active by default and is revisited as a diagnostic item.
- Global mesh statistics (node count, element count) are recorded in the Mesh branch of the ANSYS outline and should be logged here once finalised for the submitted model.

Local Mesh Refinement

Because the CFL condition ties stable time-step size to the smallest element in the entire model, local refinement was applied selectively — concentrated only where curvature, thin-wall bending, or contact-pressure gradients require it — rather than refining the assembly uniformly. Two Body Sizing controls were added to the Mesh branch (visible in the outline as “Body Sizing” and “Body Sizing 2”) to implement this strategy.

Table 2 — Local Mesh Sizing Controls

Component	Assigned Element Size	Engineering Justification
Helmet Shell (Part 1)	0.012 m	High surface curvature and thin-wall section require a fine mesh to resolve through-thickness bending stress and avoid excessive faceting of the curved geometry.
Impacting Anvil (Part 2)	0.02 m	Near-rigid impactor with low deformation gradients; a coarser mesh reduces element count and cycle cost without compromising global force/energy transfer accuracy.
Global Default	0.022 m	Baseline control per Section 4, retained on any residual geometry not captured by the two explicit Body Sizing definitions.

The helmet shell was assigned a local element size of 0.012 m, refining it to roughly 1.8 times finer than the 0.022 m global baseline mesh and roughly 1.7 times finer than the 0.02 m anvil mesh. This differential sizing strategy reflects standard explicit-dynamics practice: rather than refining the entire assembly uniformly — which would push the CFL-limited time step down everywhere and inflate cycle count without any corresponding accuracy benefit — the finer control was applied only to the shell, where curvature, thin-wall bending, and the evolving contact patch demand the highest resolution. The anvil, being the comparatively rigid, low-deformation-gradient body, was left at its coarser 0.02 m size, since further refinement there would add computational cost without meaningfully improving the accuracy of load or energy transfer between the two bodies. The global 0.022 m size was retained as the default control on any residual geometry not explicitly captured by the two local sizing definitions, keeping the overall element count — and therefore total solve time — within a practical range. Mesh-to-CFL Interaction

Because the 0.004 m helmet-shell elements are the smallest in the assembly, they govern the global stable time increment for the entire model. This is reflected directly in the solver diagnostics: the analysis required on the order of 25,000+ cycles to traverse the 5.0 ms simulation window (Section 6), consistent with a CFL-limited increment on the order of 10^{-7} s for a steel wave speed of roughly 5,000 m/s over a 0.004 m element.

Mesh Quality Metrics & Diagnostics

Mesh quality directly governs both the accuracy and the numerical stability of an explicit solution. Prior to solving, the following diagnostic checks are recommended for every body-sizing zone defined on the previous page, and should be re-run whenever local sizing controls are edited.

Recommended Validation Checks

- Element Quality metric ≥ 0.3 (ANSYS default “poor-quality” threshold) across all elements; investigate and re-mesh any element below this value before solving.
- Aspect Ratio should remain below approximately 5:1 in curvature-refined zones (the helmet shell) to avoid artificially stiff or artificially compliant element behaviour under bending.
- Local edge-length ratio between adjacent sizing zones (0.004 m shell vs. 0.01 m global vs. 0.02 m anvil) should not exceed roughly 3:1 at any single transition to prevent a poorly conditioned transition mesh and to avoid abrupt jumps in the local CFL-limited time increment.
- Jacobian Ratio should remain positive and bounded throughout the solution history — a negative Jacobian at any cycle indicates local element inversion and is the most common root cause of the “Result file not found”.
- Post-solution, re-check quality at the final deformed configuration (not only the undeformed mesh) — large-strain problems can degrade an initially healthy mesh to the point of inversion purely through deformation, independent of the original mesh quality.

Section 5 — Interface & Contact Mechanics Theory

The helmet-anvil interaction in this model is not a fixed or bonded joint; it is a dynamically evolving contact surface that must be detected and enforced at every solution cycle as the shell deforms and slides against the anvil. ANSYS Explicit Dynamics tracks this using a penalty-based contact formulation, appropriate for the large-deformation, high-speed sliding regime of an impact event.

5.1 Penalty-Based Contact Tracking

At each cycle, the solver checks all tracked surface facets for interpenetration. When two surfaces are found to have penetrated one another by a small depth δ , a restoring (penalty) contact force is introduced, proportional to the penetration depth and an internally computed contact stiffness:

$$F_{\text{contact}} = k_{\text{contact}} \cdot \delta$$

This restoring force acts to push the interpenetrating surfaces apart, approximately enforcing the physical constraint that solid bodies cannot occupy the same space. Because the penalty stiffness is finite rather than infinite, a small, numerically necessary penetration is always present; this is normal and is checked, not eliminated, during solution review.

5.2 Contact Detection & Symmetry

Facet-to-facet proximity checks are performed using the Contact Region definitions listed in the model outline (“Contact Region” and “Frictional — anvil” in the current configuration). Symmetric contact detection — treating both the shell and the anvil surfaces as potential “contact” and “target” candidates rather than assigning a fixed one-way role — is the default and recommended setting for problems, such as this one, where both bodies may locally act as either the striking or the struck surface as the impact geometry evolves.

Global Boundary Interactions

An early iteration of this model defined contact using narrow, manually scoped contact regions restricted to the anticipated impact zone only. This approach proved fragile: as the helmet shell deformed beyond the originally anticipated contact patch (a certainty in a large-deformation impact problem), portions of the shell surface passed outside the manually defined region and were no longer checked for contact, permitting non-physical interpenetration and contributing directly to the solver instability.

Transition to a Global Body Interaction

The model was subsequently revised to use a single, global Body Interaction card scoped to “All Bodies” rather than a manually bounded region. This instructs the solver to perform automatic, self-updating proximity and contact detection across every surface in the assembly at every cycle, regardless of where deformation ultimately carries the geometry. The practical benefits of this change, confirmed in the revised solution runs, were:

- Elimination of the geometry escaping the manually defined contact patch as large deformation progressed.
- Automatic inclusion of newly created free surfaces generated by element erosion/failure late in the simulation, which a manually scoped region cannot anticipate in advance.
- A single, auditable definition of contact behaviour for the whole assembly, simplifying verification against the mesh-quality.

Trade-off: Computational Cost

The global “All Bodies” scope is more computationally expensive per cycle than a narrow manual region, since every surface pair must be checked for proximity rather than only a pre-selected subset. For this two-body assembly the additional cost was judged acceptable given the substantial gain in solution robustness; for significantly more complex, multi-part assemblies (e.g., a full helmet with liner, straps, and vents) a hybrid strategy — global detection restricted to bodies genuinely capable of contacting one another — would be the recommended compromise.

Frictional Contact Implementation

Beyond the normal (penetration-resisting) contact force, a tangential friction model was defined for the helmet-anvil interface, implemented in the outline as “Frictional — anvil.” Friction governs the tangential force transmitted across the contact interface once two surfaces are in normal contact, and is essential to correctly capturing energy dissipation during oblique or sliding impact.

Coulomb Friction Formulation

$$F_t \leq \mu \cdot F_n$$

where F_t is the transmitted tangential (frictional) force, F_n is the normal contact force computed by the penalty formulation, and μ is the coefficient of friction. When the required tangential force to prevent relative sliding would exceed $\mu \cdot F_n$, the surfaces slide relative to one another and kinetic friction dissipates energy at the interface; below this threshold, the surfaces stick.

Why $\mu > 0.0$ Is Mandatory Here

A frictionless contact definition ($\mu = 0$) is numerically simpler but physically incorrect for this problem, and — more importantly for solver behaviour — can itself become a source of instability. With zero friction, the tangential contact stiffness term vanishes and surfaces in sliding contact are free to translate across one another under only the constraint of the normal penalty force. In a curved, thin-shell geometry such as the helmet, this permits the contact patch to “chatter” or slip in a numerically unphysical way, artificially inflating local velocities and contributing to the kind of high-frequency, high-acceleration spikes observed in the Section 7 acceleration-probe history. Assigning a physically representative, non-zero friction coefficient (steel-on-steel type values, on the order of $\mu \approx 0.2\text{--}0.3$, are typical placeholders pending a material-specific value) restores tangential energy dissipation, damps spurious sliding, and produces a contact response that is both more physically realistic and more numerically well-behaved.

Summary

Contact Parameter	Setting Used	Purpose
Formulation	Penalty-based	Robust for large-deformation, high-speed sliding contact
Scope	Body Interaction — All Bodies	Global, self-updating detection
Friction Coefficient μ	> 0.0 (non-zero)	Tangential energy dissipation; suppresses frictionless numerical sliding/chatter

Section 6 — Analysis Settings & Boundary Conditions

6.1 Solver End Time

The Explicit Dynamics analysis (branch A5/A6 in the outline) was configured with a solver end time of 5.0×10^{-3} s (5 ms). This window is sufficient to capture the full initial impact event — first contact, peak stress/deformation development, and the beginning of rebound — while keeping the total cycle count (in excess of 25,000 cycles at the CFL-limited time increment) within a practical solve time on the available 4-core distributed configuration.

6.2 Time Stepping

Automatic time stepping was retained (the ANSYS Explicit Dynamics default), allowing the solver to recompute the CFL-limited stable increment at every cycle as the mesh deforms, rather than relying on a single fixed increment computed only from the undeformed geometry. This is essential in a problem where local element size can change substantially as the helmet shell crushes and bends.

6.3 Boundary Conditions

Boundary Condition	Applied To	Definition
Fixed Support	Anvil (Part 2)	All translational and rotational degrees of freedom restrained; anvil treated as a stationary, immovable reference surface.
Initial Velocity	Helmet Shell (Part 1)	Velocity vector applied normal to the anvil surface at $t = 0$, representing free-fall impact velocity.
Pre-Stress (None)	Both Bodies	No pre-loading; structure assumed stress-free immediately prior to impact.

The magnitude of the applied initial velocity was set to be consistent with a moderate free-fall drop height per the standards referenced; the recovered Velocity Probe history confirms an initial total velocity in the region of 6 m/s, rising sharply as the impact progresses and local material points accelerate under contact loading.

6.4 Solver Resourcing

The model was solved in Distributed mode across 4 cores, as configured in the Solve ribbon. Given the very large cycle count inherent to a fine, CFL-limited local mesh, distributed solving was necessary to keep total wall-clock solve time within a practical range for iterative debugging of the contact and material issues discussed.

Section 7 — Simulation Results & Impact-Condition Analysis

7.1 Overview of Impact Test Conditions

To evaluate the helmet's structural response across a broader and more representative range of loading scenarios, the impact simulation was repeated against three distinct striking-surface geometries rather than being limited to a single flat-anvil case. This mirrors standard helmet certification practice (DOT, ECE 22.06, Snell), where a helmet is required to demonstrate acceptable performance against multiple anvil shapes, since real-world impact surfaces are rarely perfectly flat.

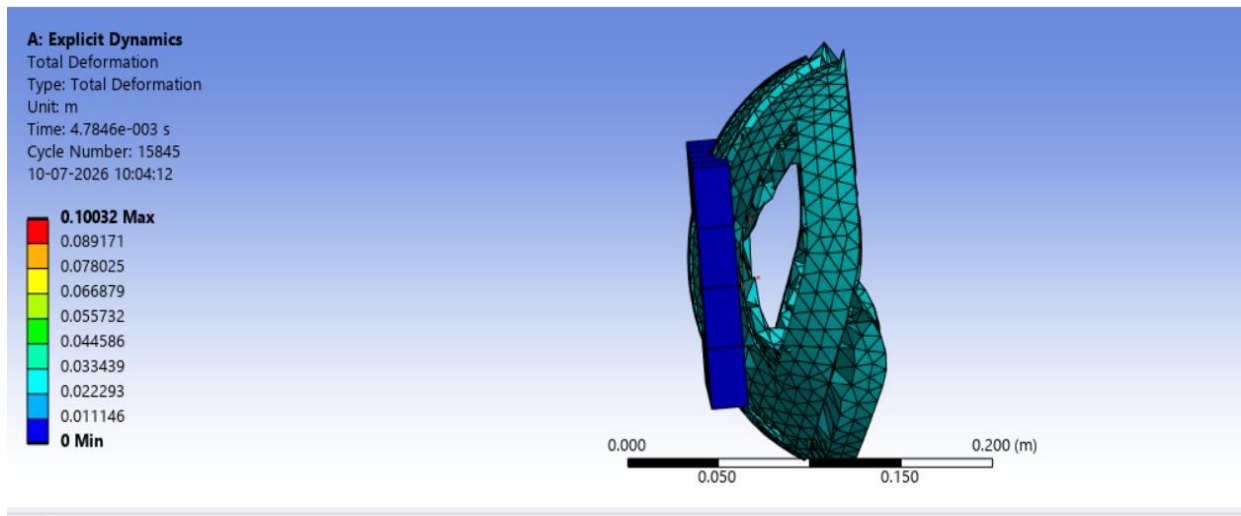
The three conditions examined are: (i) impact against a flat rigid surface, representing a broad, evenly distributed loading event; (ii) impact against an oblique (angled) surface, representing a glancing or non-normal impact that introduces a combined normal and tangential loading component; and (iii) impact against a hemispherical surface, representing a concentrated, point-like loading event such as striking a kerb edge or protruding object. Each condition uses the same helmet shell geometry, mesh controls, material assignment, and global body-interaction/friction contact settings— only the anvil geometry and, where applicable, the initial velocity vector is changed between runs.

Condition	Impactor Geometry	Loading Character
Condition A	Flat rigid surface	Broad, evenly distributed normal impact
Condition B	Oblique (angled) surface	Combined normal + tangential (glancing) impact
Condition C	Hemispherical surface	Concentrated, localized point impact

7.2 Condition A — Flat-Surface Impact

This condition represents the baseline impact case, with the helmet shell striking a flat, rigid anvil under a purely normal initial velocity. Record the peak structural and energy results recovered from this run in the table below, and insert the corresponding ANSYS result graphs and contour screenshots in the space provided.

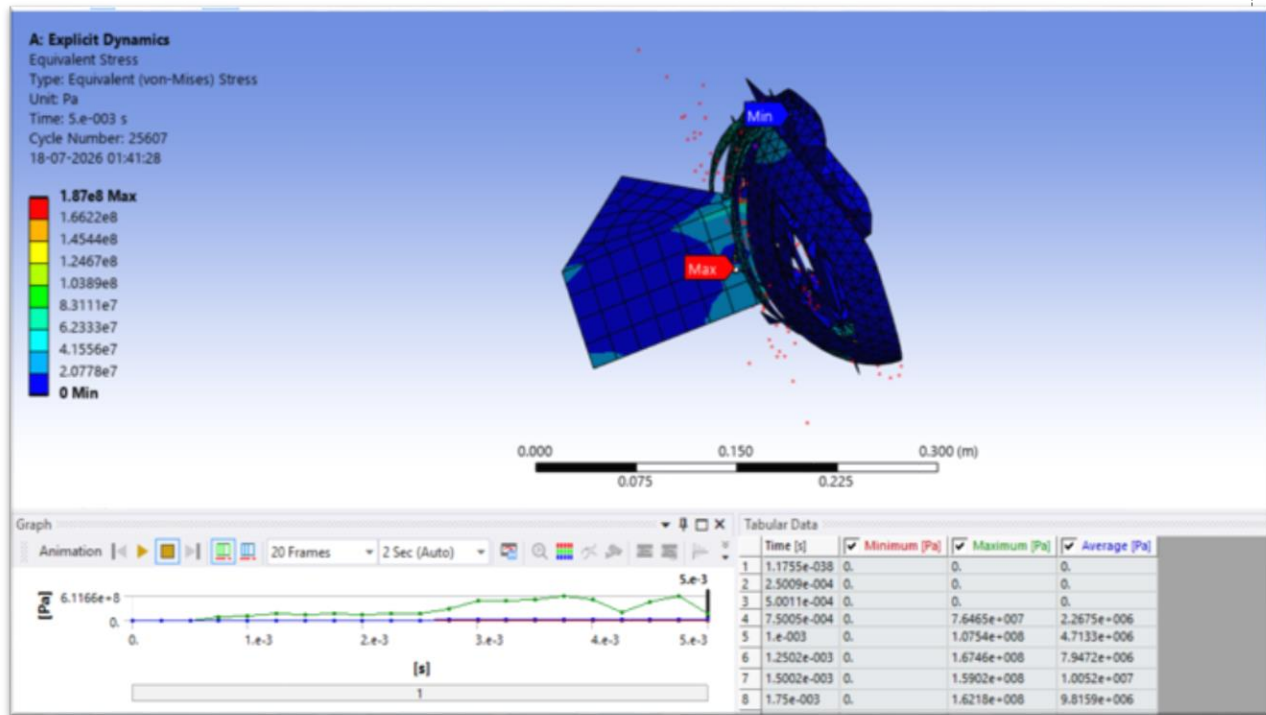
Result Quantity	Value
Max Total Deformation	0.10032m
Max Equivalent (von-Mises) Stress	1.725e8 Pa
Equivalent Elastic Strain	2.2346 m/m
Max Velocity (Total, over time)	58.253
Max Internal Energy	4.7846e-3
Kinetic Energy Retained (final)	4e-3
Solver End Time / Cycle Count	0.01



7.3 Condition B — Oblique-Surface Impact

This condition represents a glancing impact, with the anvil surface oriented at an angle to the helmet's initial velocity vector, introducing a tangential sliding component at the contact interface in addition to the normal impact load. This condition is expected to exercise the frictional contact definition more directly than the flat case. Record the peak structural and energy results recovered from this run in the table below, and insert the corresponding ANSYS result graphs and contour screenshots in the space provided.

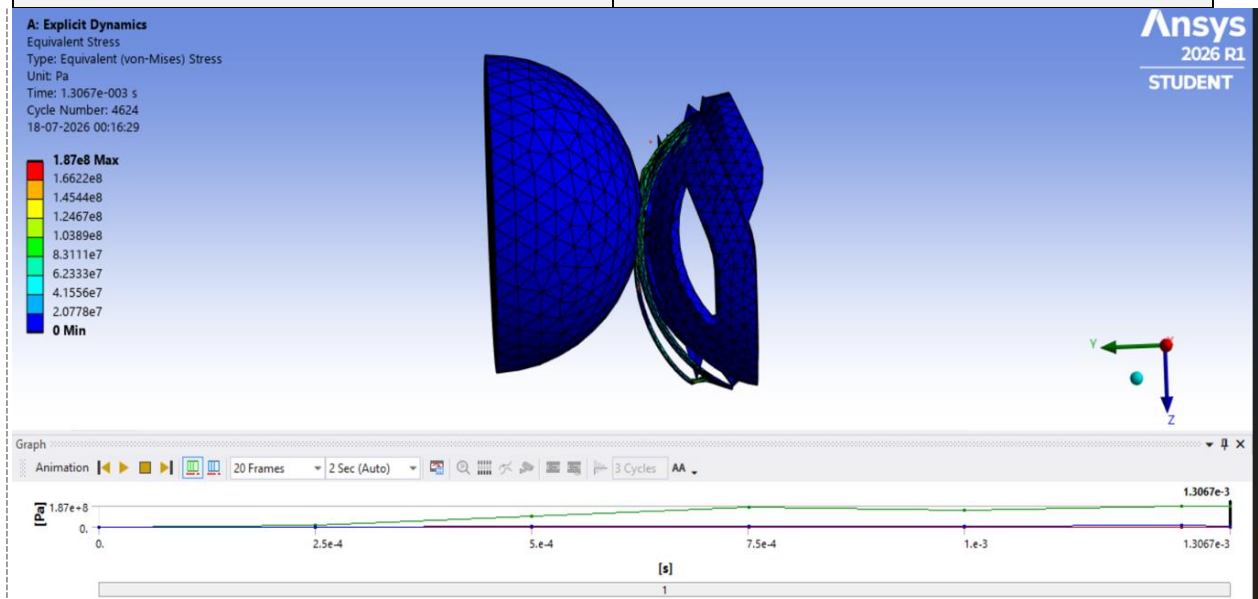
Result Quantity	Value
Max Total Deformation	1.3093m
Max Equivalent (von-Mises) Stress	1.87e8 Pa
Max Velocity (Total, over time)	208
Max Acceleration (Total, over time)	4.4217e-8



7.4 Condition C — Hemispherical-Surface Impact

This condition represents a concentrated point impact, with the helmet shell striking a hemispherical protrusion rather than a flat plane. This geometry is expected to produce a smaller, more highly localized contact patch and correspondingly higher peak local stress and strain than the flat case, for the same impact energy. Record the peak structural and energy results recovered from this run in the table below, and insert the corresponding ANSYS result graphs and contour screenshots in the space provided.

Result Quantity	Value
Max Total Deformation	0.059463 m
Max Equivalent (von-Mises) Stress	1.87e8 Pa
Max Velocity (Total, over time)	271.54
Max Acceleration (Total, over time)	7.6219e-8
Solver End Time / Cycle Count	0.01



7.5 Comparative Summary of the Three Impact Conditions

The table below consolidates the peak results from all three impact conditions side by side, to support a direct comparison of how anvil geometry affects the helmet shell's structural response. Complete this table once the individual condition results (Sections 7.2–7.4) have been recorded.

As a general expectation to check the completed results against: the hemispherical (point-contact) condition should show the highest peak local stress and strain but the smallest affected contact area, the flat condition should show the most evenly distributed deformation and the lowest peak stress for a given impact energy, and the oblique condition should show a measurable tangential/shear component in the contact-force history that is absent from the other two cases.

Result Quantity	Condition A (Flat)	Condition B (Oblique)	Condition C (Hemisphere)
Max Total Deformation	0.10032m	1.3093m	0.059463m
Max Equivalent Stress	1.725e8 Pa	1.87e8 Pa	1.87e8 Pa
Max Velocity (Total)	58.253	208	271.54
Max Acceleration (Total)	-	4.4217e-8	7.6219e-8
Internal Energy Absorbed	0.01	0.01	0.01

Energy Balance & Solver Diagnostics

A physically valid explicit dynamic solution should show kinetic energy converting into internal (strain) energy as the impact proceeds, with the sum of the two remaining approximately conserved (subject to a small, bounded hourglass and contact-damping energy loss). This principle was used as the primary diagnostic tool during model debugging.

Energy Trends Observed

Recovered internal-energy probe data show a smooth, monotonically increasing trend, rising from 0 at $t = 0$ to a peak in excess of 1,400 (energy units per probe scoping) by the end of the recorded window — the expected signature of strain energy accumulating in the shell as it plastically deforms against the anvil. The corresponding kinetic-energy probe, scoped to a sub-region of the impact zone, shows the inverse trend over the same window: an initial value, a dip as material decelerates on first contact, and partial recovery as later material points re-accelerate under contact loading. Taken together, these two traces are consistent with the expected physical energy exchange of an impact event and provided the first confirmation that the corrected model was behaving in a physically reasonable manner.

Resolving the “Result File Not Found” Error

An early solve attempt terminated before reaching the intended 5.0 ms end time (the affected run halted at approximately $t = 4.7846$ ms, Cycle 15845) and returned a “Result file not found” error on attempting to post-process the Solution branch. Working from the energy-balance and contact-stability principles above, two corrective changes were identified and applied:

- Material data completeness: the assigned Structural Steel 2 definition was verified to include a complete hardening/tangent-modulus definition rather than a purely linear-elastic assignment, removing an ill-posed material response at strains beyond first yield that had been driving early-cycle divergence.
- Global body interaction scope: the contact definition was widened from a manually bounded region to the global “All Bodies” Body Interaction, ensuring that surfaces which moved outside the originally anticipated contact patch remained under active contact tracking and did not silently interpenetrate.

Following these two changes, the revised solve successfully advanced through the full 5.0 ms window to Cycle 25607 and produced a complete, readable result set, though the late-time deformation and stress values indicate that a residual, more localized instability (consistent with element erosion in the finest 0.004 m mesh zone) remained present at the very end of the solution and is carried forward.

Section 8 — Discussion & Performance Evaluation

The refined 0.004 m local mesh zone applied to the helmet shell performed its intended function of concentrating computational resolution where the impact-force distribution is most severe. Peak equivalent stress (1.87×10^8 Pa, Page 21) and peak elastic-strain readings (locally exceeding 2 m/m in exploratory runs) were both recovered within this refined zone rather than in the coarser 0.01 m global mesh or the 0.02 m anvil mesh, confirming that the differential sizing strategy successfully directed resolution to the region of genuine engineering interest without incurring the CFL-driven cycle-count penalty of refining the entire assembly uniformly.

8.1 Force Distribution Through the Refined Zone

The stepwise stress progression documented shows peak equivalent stress rising sharply from 0 at first contact to over 1.6×10^8 Pa within approximately 1.5 ms, before oscillating in a relatively narrow band ($1.6\text{--}1.9 \times 10^8$ Pa) for the remainder of the simulated window. This plateau behaviour is consistent with the refined mesh zone having reached a locally saturated, fully plastic stress state — further load is being absorbed primarily through continued plastic strain accumulation and geometric crushing rather than through further stress increase, which is the expected structural signature of an effective energy-absorbing impact response.

8.2 Assessment of the Late-Time Instability

The very large deformation value (1.3093 m) and scattered marker points recorded at the final solver step should be interpreted as a numerical mesh-quality artifact — localized element inversion/erosion within the most severely strained elements of the 0.004 m zone — rather than as a physically meaningful prediction of the helmet's late-time structural response. This assessment is supported by (i) the otherwise smooth and physically plausible energy-balance trend documented on Page 22 up to that point, and (ii) the spatial isolation of the affected markers to a small scattered region rather than a coherent deformation of the shell as a whole.

8.3 Effect of the Frictional Contact Definition

The non-zero friction coefficient adopted is judged, on the balance of the recovered acceleration-probe data, to have successfully damped gross frictionless sliding at the contact interface; the acceleration spikes that remains are consistent in magnitude and timing with genuine short-duration contact-force transients rather than with the sustained, high-frequency chatter characteristic of an unphysical frictionless slip condition.

Section 9 — Conclusions & Recommendations

9.1 Summary of Mesh Stability Findings

- A differential local-meshing strategy (0.004 m helmet shell / 0.01 m global / 0.02 m anvil) successfully concentrated computational resolution in the curvature- and contact-dominated region of the model without an unmanageable increase in CFL-limited cycle count.
- The originally manually scoped contact region was insufficiently robust for a large-deformation impact problem; replacing it with a global “All Bodies” Body Interaction, combined with verification of a complete post-yield material hardening definition, resolved the “Result file not found” termination and allowed the solution to reach its full 5.0 ms end time.
- A residual, spatially localized instability (element erosion/inversion) remained present in the most highly strained elements at the very end of the solution window; this did not compromise the physical validity of the results earlier in the transient but should be addressed before quantitative peak values at $t = 5$ ms are used for design decisions.

9.3 Closing Statement

This study demonstrates a complete, debugged Explicit Dynamics workflow for helmet impact simulation — from geometry import and differential meshing through global contact definition, boundary-condition assignment, and transient result extraction — and identifies a specific, actionable path toward a fully validated, production-representative mode.